Enphase Energy

ENPH | NASDAQ | 2026-10-07

Q4 2027 model ceiling: USD 13.63

Analytical Base 38/60 = 63 | Technology optionality / Watchlist

20x forward-four-quarter normalized GAAP operating profit including SBC and D&A, less outstanding note principal at labeled quarter end, divided by end-window economic shares. No cash addition. Quarter includes itself plus next three. Apply the common-equity floor inside each of 24 operating outcomes before weighting; all outcomes are cash-funded.

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

All operating values are in USD m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Writing-date snapshot: USD 33.91 per share (2026-10-06 close 16:00 EDT); 132.145 million common shares (2026-07-20); estimated equity capitalization USD 4,481.0 million. Dated exchange quote from Stock Analysis; legal common shares from Q2 10-Q cover. Not the model economic dilution count. This dated common-share snapshot is separate from forecast diluted shares.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	124.1	20x	139.900	13.63

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

Consolidated operating profit; tariff-refund one-off removed and manufacturing credits shown separately.. 20x forward-four-quarter normalized GAAP operating profit including SBC and D&A, less outstanding note principal at labeled quarter end, divided by end-window economic shares. No cash addition. Quarter includes itself plus next three. Apply the common-equity floor inside each of 24 operating outcomes before weighting; all outcomes are cash-funded.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use USD m unless a different unit is stated.

Thesis

Enphase has a credible route from distributed solar electronics into AI data-center power conversion. Its proprietary control silicon, high-frequency magnetics and bidirectional GaN architecture are relevant engineering assets. Commercial SST revenue, however, comes after a period of full-rack qualification and customer pilots. The investment case therefore depends first on whether the existing solar and battery business can recover while retaining margins after price cuts, financing disruption and the eventual reduction in manufacturing credits.

The active Q4 2027 ceiling is **$13.63 per economic share**, using 20x forward-four-quarter normalized GAAP operating profit, deducting the $575 million note principal still outstanding at that quarter end and dividing by 139.900 million end-window economic shares. The window is Q4 2027 through Q3 2028. It includes SBC, depreciation, amortization and ordinary warranty expense. Cash is used to test funding and debt repayment; it is not added to this operating-profit valuation. This is a company-derived ceiling, not a consensus target or a valuation inferred from a semiconductor or AI multiple.

The common-equity floor is applied separately inside each of 24 coherent outcomes. It adds only 1.81 cents and 0.92 cents to the first two quarterly ceilings versus flooring the average, and leaves all later ceilings unchanged. Negative conditional operating profit and raw common value remain visible.

The analytical Base is **63/100**. The engineering platform deserves research, but residential financing, policy-dependent profit, competition and unproven rack qualification limit the score. Covered status does not establish that the stock is held.

What Enphase sells and what customers pay for

The established business combines module-level microinverters, battery systems, control hardware, monitoring and related energy products. A microinverter converts a solar module's output locally; distributed conversion and monitoring can improve installation flexibility, fault isolation and serviceability. Batteries provide backup, self-consumption and time-of-use management. Meter-collar approval, installer training and a supported installed base can matter as much as a component efficiency claim.

Enphase reported $1.473 billion of 2025 sales, approximately 6.4 million microinverters and 706.1 MWh of batteries. The United States supplied about 81% of sales and one customer accounted for 39%. Contract manufacturers such as Flex, Salcomp and Sunwoda provide manufacturing capacity. The business is not a diversified domestic factory operator immune to distributor or consumer-financing stress. 2025 Form 10-K

SolarEdge, Tesla, Huawei, Sungrow and other inverter or storage competitors constrain pricing. Battery economics also compete with cheaper DC-coupled installations and alternative uses of household capital. Enphase's AC-coupled retrofit opportunity is useful because it can address existing solar sites, including systems with other vendors' inverters, but it does not guarantee premium pricing. The Q1/Q2 calls describe actual price

adjustments in Europe and the United States. Cost reductions must first offset those concessions and customer acquisition costs.

The fifth-generation battery is intended to reduce cost per kWh by roughly 40% and improve energy density by 50%. These are product targets and comparisons, not a 40-point consolidated margin uplift. Some savings are passed to customers; the model lowers blended battery revenue per kWh from $500 in its Q3 calibration to $450 in 2027, with further declines thereafter. IQ Vault 80 adds an 80 kWh commercial cabinet and is expected to begin shipments in Q1 2027. Commercial solar and storage remain nested within the existing product vectors, not an additional full-market revenue pool.

The latest earnings require a careful normalization

Q2 2026 revenue was $291.854 million, with 1.59 million microinverters and 113.8 MWh of batteries. Reported gross profit was $175.014 million and GAAP operating profit was $51.519 million. The result included **$45.4 million of tariff-refund benefit in gross profit**. Removing that one-off leaves $129.614 million of gross profit and **$6.119 million of operating profit**. This establishes a much lower recurring starting point than the reported 60% gross margin suggests. Q2 filing

This normalization differs from management's non-GAAP reconciliation: the latter removes $43.4 million of prior-period refund and also excludes compensation and other expenses. Here the full $45.4 million is removed from recurring profit, while the $43.554 million of quarterly SBC remains an expense. The refund is not then subtracted again from future margins. The $11 million received in July is a cash collection of a previously recognized receivable, not new forecast profit. Q1's $18.905 million charge associated with selling prior-year manufacturing credits is likewise not charged repeatedly.

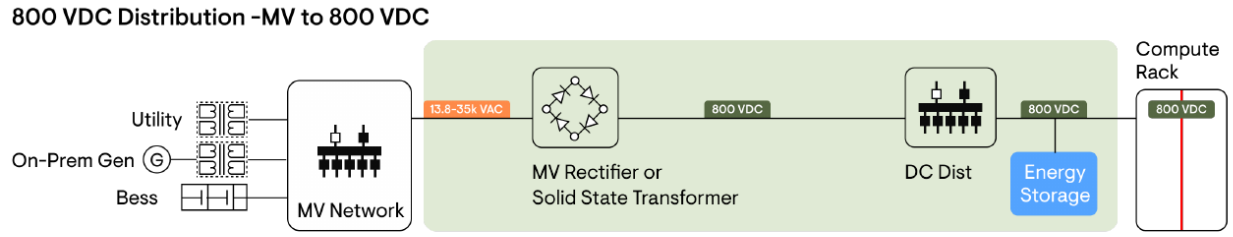
Q2 U.S. sell-through fell 34% year over year and 7% sequentially. Europe improved, but the geographic mix and distribution inventory explain why a recovery should not be assumed to arrive uniformly. Q3 guidance is revenue of $290-320 million, battery shipments of 130-150 MWh, GAAP gross margin of 42-45%, and GAAP operating expense of $120-124 million. The reference midpoint is $305 million revenue, $132.675 million GP and $10.675 million OP. The model does not substitute non-GAAP expense guidance for this fully expensed starting point. Q2 release

Management's Q4 comment covers $61.2 million of safe-harbor sales; it is not total Q4 guidance. The $315 million total Q4 reference is an analytical assumption. The broader $1.081 billion of signed safe-harbor arrangements includes $878.6 million under the physical-work-test method, with shipments expected mainly from 2028 through 2031. These amounts are already represented within modeled units. They are not added to ordinary sales, and customer payments are not treated as costless financing or profit.

Why the SST architecture is technically interesting

An SST substitutes switched power electronics and high-frequency magnetic conversion for part of the conventional transformer-and-conversion chain. Enphase aims to convert medium-voltage AC directly to the 800 V DC architecture being developed for dense AI racks. The potential benefits are compact magnetics, rapid control and fewer conversion stages. The customer purchases reliable delivered power and system availability, rather than a laboratory switching frequency.

The shift to 800 VDC architecture



Source: NVIDIA white paper "800 VDC Architecture for Next-Generation AI Infrastructure" by Jared Huntington & Mike Tu

Opportunity

- Convert MV AC to 800 VDC and deliver directly to GPU rack
- Reduce complexity by eliminating multi-stage power conversion
- Eliminate the sidecar and move energy storage near the SST

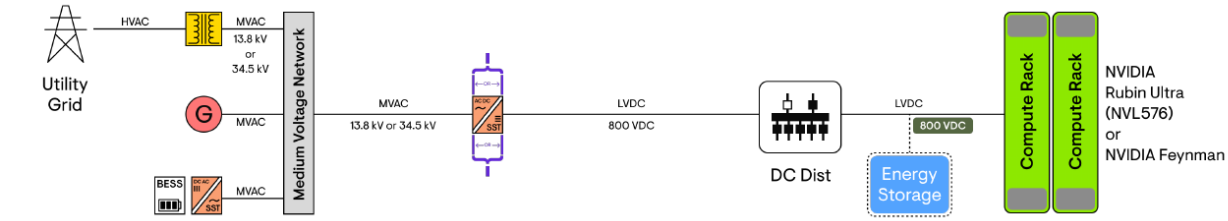
ENPHASE.

© 2026 Enphase Energy, Inc. 6

Issuer 800 V architecture; under development, not proof of customer acceptance.

The architecture is relevant to rapid AI load changes. Enphase proposes moving energy buffering toward a central battery system where its control response allows it, while also supporting customers who retain local high-rate batteries. Neither design eliminates energy storage from the physical system. A second SST for the BESS would be additional hardware only if a customer adopts that topology; it is not automatically twice the wallet for every compute megawatt.

The IQ SST solution: supports two use cases



Enphase's differentiated solution: SST coupled with a BESS in the medium voltage section, and an SST for the compute load in the low voltage section

Enphase differentiated solution

- IQ SST supports two use cases:
  1. **Without LV battery:** Sub-millisecond response time enables us to reflect rapid compute load changes to the BESS SST
  2. **With LV battery:** A DC/DC block to charge and discharge a LV battery which is part of the IQ SST skid
- Reliability: Modular architecture enables very high reliability to achieve 99.999% uptime
- Manufacturing scale and cost: Soft switching and widely available components drive lower cost; supplied from U.S. manufacturing facilities

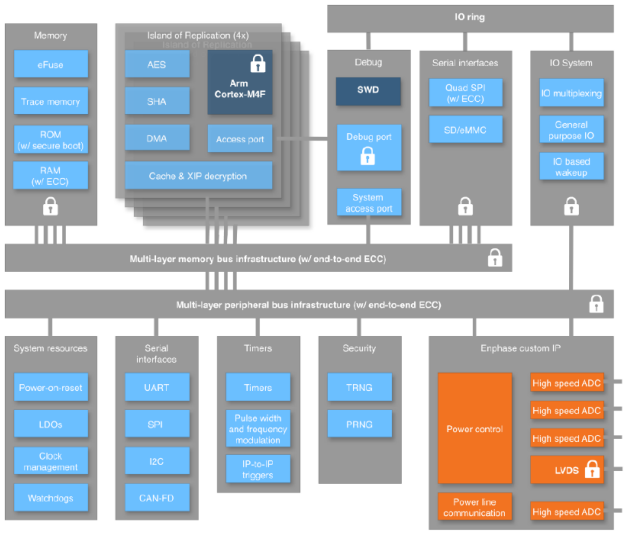
ENPHASE. Product under development. Specifications and performance subject to change.

© 2026 Enphase Energy, Inc. 7

Central BESS and local battery use cases are alternatives; both require system qualification.

Kestrel is Enphase's fifth-generation proprietary ASIC, fabricated on 22 nm CMOS. Its four processor islands separate software functions, while a dedicated hardware pipeline handles the fast converter-control path. The whitepaper describes four 100 Msps ADCs and control bandwidth above 100 kHz. Reuse across solar, batteries, EV charging and SST can spread engineering costs across multiple products. The economic advantage is reusable control IP and integration; it does not require leading-edge logic economics.

- Operational longevity: The architecture is designed with significant performance margins to accommodate component derating, maximizing long-term reliability in demanding environments.



Enphase Energy Kestrel ASIC architecture

The ASIC utilizes a multi-processor architecture featuring four identical islands of replication (IoR). Each IoR is equipped with an ARM Cortex-M4F processor, a dedicated DMA controller, and integrated cryptography accelerators.

By employing a defined software convention, each island is assigned a distinct, isolated role. In a microinverter product, the functional distribution is structured as follows:

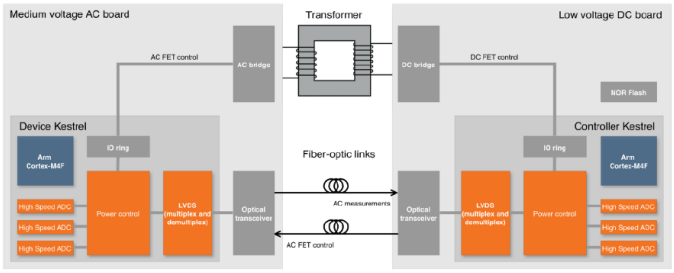
- Island 0, Secure Boot: Manages hardware-rooted trust and cryptographic integrity. This island effectively implements a secure enclave within the chip.

© 2026 Enphase Energy, Inc. · All rights reserved. Page 6

Original Kestrel architecture; reusable silicon can reduce development cost but does not establish rack reliability.

The SST module uses two Kestrel chips on opposite sides of the isolation barrier. Fiber links carry measurements and control commands; a high-frequency transformer transfers the power. This distinction matters: an optical control link is not evidence of a large datacenter transceiver sales award, and magnetic insulation must still pass its own medium-voltage tests.

ENPHASE.



IQ SST power module — tandem configuration of two Kestrel chips

Within the tandem configuration, the low-voltage domain Kestrel chip will take the lead in the power conversion. This chip is identified as the “Controller Kestrel,” whereas the other chip is identified as the “Device Kestrel.” The Device Kestrel conveys AC power domain measurements to the Controller Kestrel. The Controller Kestrel conveys AC FET control to the Device Kestrel.

LVDS Controller

The fiber-optic isolation link between the two Kestrel ASICs in the IQ SST power module is driven by the Kestrel's custom LVDS controller IP. This LVDS controller interfaces with an external optical transceiver chip and serializes and deserializes multiple logical power-control streams into a single LVDS bitstream as illustrated below.

Controller to Device Kestrel	Device to Controller Kestrel
AC FET control information	AC power domain measurements
Remote shutdown request	Remote shutdown request
Remote chip access	Remote chip access

Logical power-control streams multiplexed into LVDS bitstream

The remote chip access enables the Controller Kestrel to access the 32-bit address space of the Device Kestrel and vice versa. The LVDS controller includes a hardware protection unit to restrict these accesses. The Controller Kestrel uses remote access to upload code in the Device Kestrel internal RAM, eliminating the need for an external NOR Flash chip on the medium-voltage board.

The external optical transceiver chip requires the LVDS bitstream to be DC balanced. The LVDS controller accomplishes this through scrambling and 6b/8b coding. Latency-critical streams are scrambled to minimize transmission delays. Latency-tolerant streams use 6b/8b coding, which ensures DC balance and provides built-in error detection capabilities.

© 2026 Enphase Energy, Inc. · All rights reserved. Page 16

Fiber isolation carries measurements and control between two ASICs; it is distinct from magnetic power isolation.

The issuer targets roughly 98.5% peak conversion efficiency, sub-millisecond response and five-nines availability. These are separate measures. A fast response target does not prove lifetime availability, partial-discharge performance or fleet efficiency. The whitepaper's approximate thousandfold speed comparison is not established by simply dividing “tens of milliseconds” by “less than one millisecond”; the model uses the absolute target and requires customer-level validation.

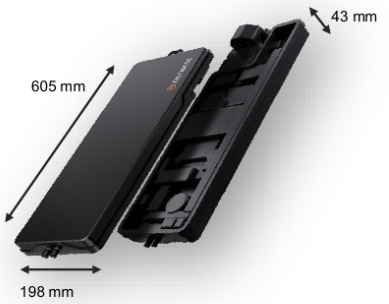
Manufacturing repetition is useful, but qualified output is the constraint

The July deck describes a 4 kW module and a 2.5 MW rack containing 672 modules. The May whitepaper described a different, 342-module arrangement. These are revisions, not interchangeable unit counts. At the stated module rating, 672x4 kW equals 2.688 MW. If 10% of those modules were unavailable, only 2.4192 MW would remain by that simple arithmetic. Final redundancy and rating conventions therefore require clarification before claiming the full 2.5 MW survives a specified module-loss fraction.

Our distributed architecture

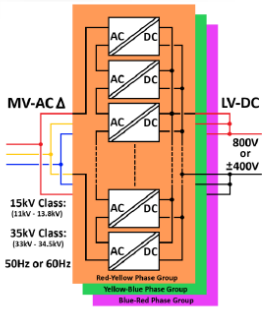
Enphase power module

- 4 kW building block; over 98.5% peak efficiency expected
- 250 – 500 kHz switching frequency; control ASIC: Kestrel (proprietary Enphase)
- BDS and UDS GaN for input and output power stage
- Weight: ~8 lbs; plastic enclosure with silicone potting
- Soft-switching design for low EMI



How power modules create an IQ SST rack

- Multiple power modules stacked depending on MV input and power
- Series connection on the input; parallel connection on the output (ISOP)
- Delta configuration; no neutral
- Input has triple redundant short-on-fail circuit (anti-fuse)
- Dual sided backplane for power module connection
- Sub-millisecond response time
- Output has short circuit current limit (fused)



ENPHASE. Product under development. Specifications and performance subject to change.

© 2026 Enphase Energy, Inc. 11

4 kW module targets and protection architecture; peak targets are not fleet results.

IQ SST 2.5 MW rack @ 34.5 kV

- Power: 2.5 MW, ambient-air cooled
- Architecture: supercluster of 672 power modules (Δ configuration)
- Will support 35 kV and 15 kV utility interconnection classes
- Designed for five-nines availability
- 10-year limited warranty
- Modular hot-swap design — service with a cobot
- No internal batteries required
- U.S.-based manufacturing
- FEOC-compliant supply chain
- Scales up to 5 MW per skid



ENPHASE. Product under development. Specifications and performance subject to change.

© 2026 Enphase Energy, Inc. 13

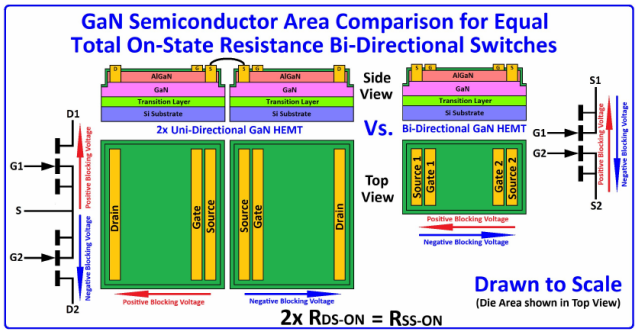
July rack revision: 2.5 MW and 672 modules; rating and redundancy arithmetic require final reconciliation.

The September 8 update says Texas power modules are being built for full-rack assembly and validation. Its testing scope includes electrical and thermal performance, efficiency, distributed control, redundancy and partial discharge. The next milestone is a full-system demonstration expected in November 2026, followed by pilots in 2027. Existing microinverter manufacturing scale is relevant process experience; it is not evidence that a

comparable number of SST modules already meet customer qualification, yield and lifetime requirements.
September issuer update

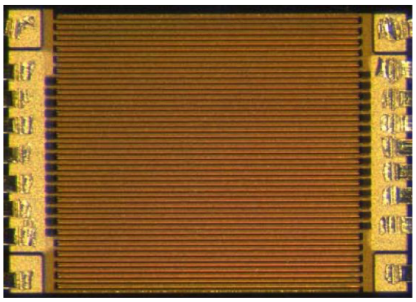
The GaN whitepaper offers a useful first-principles argument. A monolithic bidirectional switch can avoid two larger back-to-back devices at a matched conduction resistance. The cited fourfold die-area advantage is a comparison at the transistor level. Packaging, gate drivers, magnetics, isolation, assembly, testing, cooling and warranty cost remain. The model never translates this into a fourfold system margin improvement.

ENPHASE.



A 'cartoon' drawing illustrating the GaN HEMT structure and the inherent 4x die area advantage of the GaN HEMT BDS compared to an anti-series connected pair of back-to-back conventional uni-directional GaN HEMT devices.

Furthermore, because traditional back-to-back configurations place two devices in series, each uni-directional transistor must be physically upsized to counteract the doubled conduction losses. A single GaN BDS replaces both oversized components. Because it eliminates the need for two series connected devices that are each twice its size, the GaN BDS structure delivers a 4x die-area advantage over conventional back-to-back pairs [5]. This 4x reduction in semiconductor die footprint area translates directly into up to a 4x cost advantage, forming the foundational value proposition of GaN BDS technology.



A closeup view of a GaN HEMT BDS device die [6].

© 2026 Enphase Energy, Inc. · All rights reserved. Page 6

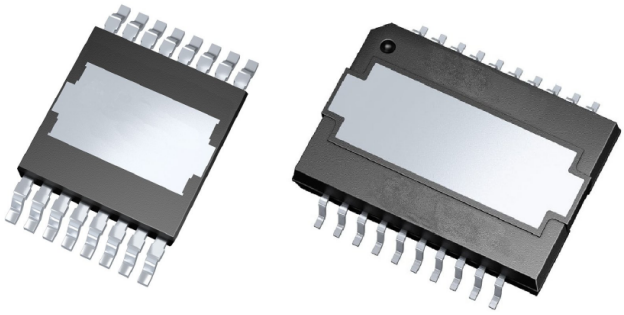
Up to fourfold die-area saving is a device comparison at matched conduction resistance, not a fourfold system cost cut.

The issuer names Infineon, Renesas and Navitas as sources of commercially available 650 V bidirectional devices, with other vendors developing products. The SST discussion separately specifies a 900 V device class and prospective qualification. A named 650 V supplier is not a disclosed 900 V SST production winner. Standardized packages also create a route to substitution and limit supplier scarcity rents.

ENPHASE.

04 Commercial Availability

Commercially available GaN HEMT Bi-Directional Switch (BDS) devices utilize industry-standard, top-side cooled surface-mount packages defined by the Joint Electron Device Engineering Council (JEDEC). This standardization allows multiple semiconductor companies to develop pin-for-pin compatible devices. This multi-supplier ecosystem can help Enphase reduce sourcing risk and support long-term platform scalability.



GaN HEMT BDS surface-mount top-side cooled packages: Left = TOLT (MO-332) & Right = DSO-20 (MO-166).

As of this writing, commercially available GaN BDS devices are offered by Infineon, Renesas, and Navitas. These initial components feature a 650V rated operational voltage and an 800V momentary surge voltage rating. They are available in industry-standardized maximum on-state resistance ($R_{SS\text{-}ON}$) values of 30mΩ, 70mΩ, and 140mΩ.

At the March 2026 Applied Power Electronics Conference (APEC) in San Antonio, Texas, STMicroelectronics and Texas Instruments announced and showcased their upcoming GaN HEMT BDS devices. These future offerings are expected to also conform to this existing industry GaN HEMT BDS standardization.

© 2026 Enphase Energy, Inc. · All rights reserved. Page 9

Issuer names several available 650 V BDS vendors; this is not an awarded 900 V SST bill of materials.

ENPHASE.

This 335VAC RMS operational requirement is 40% higher than the standard single-phase 240VAC RMS voltage level than can be reliably serviced using 650V rated power semiconductor switches. Accordingly, the IQ SST power module will make use of a new voltage class of 900V rated GaN BDS devices. This ~40% increase in the GaN BDS rated voltage matches the 40% increase in the IQ SST power module AC operational voltage.

The existing commercially available 650V rated GaN BDS devices that feature an 800V momentary surge voltage rating are fabricated using a GaN on silicon process. By increasing both the lateral and vertical GaN HEMT feature dimensions, it is possible to extend these GaN BDS designs to support a rated operational voltage of 900V, with a momentary surge voltage rating of 1.1kV. This is currently believed to be the practical economic limit for the existing GaN-on-silicon process, with development of 1.2kV GaN BDS devices being based on GaN on some form of insulative substrate such as sapphire (Al2O3) or the engineered Qromis Substrate Technology (QST) [25].

Several semiconductor companies are currently developing 900V rated variants of GaN BDS devices based on existing GaN on silicon processes. These 900V rated variants are targeted for adoption in Electric Vehicle (EV) On Board Charger (OBC) applications enabling compatibility for operation from a three-phase utility grid (380VAC to 480VAC RMS). This intended application results in devices with a maximum on-state resistance of 15mΩ being developed which are ideally sized for use in the Enphase 4kW IQ SST power module. Engineering samples of these 900V rated GaN BDS devices are currently being evaluated in the Enphase IQ SST power module and fully JEDEC-qualified production parts are expected to be available by the end of 2026.

The IQ SST power module is designed to support both the NVIDIA 800VDC as well as the Open Compute Project (OCP) ±400VDC requirements. The DC output will be produced using a series connected pair of 400VDC bridges. Each of these output bridges will make use of 650V rated GaN HEMT Uni-Directional Switch (UDS) devices. This converter topology choice using a series connected pair of DC bridges is designed to support the OCP split-rail ±400VDC configuration and provide a practical path toward future higher-voltage DC architectures such as 1,500VDC based on a ±750VDC design that is expected to use 900V GaN UDS devices.

To maximize design efficiency, semiconductor manufacturers are currently co-developing 900V GaN UDS and BDS devices using a unified platform architecture. This common design framework features limited variation between the two device derivatives, enabling expected shared development resources. Consequently, both the bi-directional and uni-directional variants can undergo parallel design and qualification.

© 2026 Enphase Energy, Inc. · All rights reserved. Page 20

May 900 V device development and qualification expectations; subsequent production status remains a catalyst.

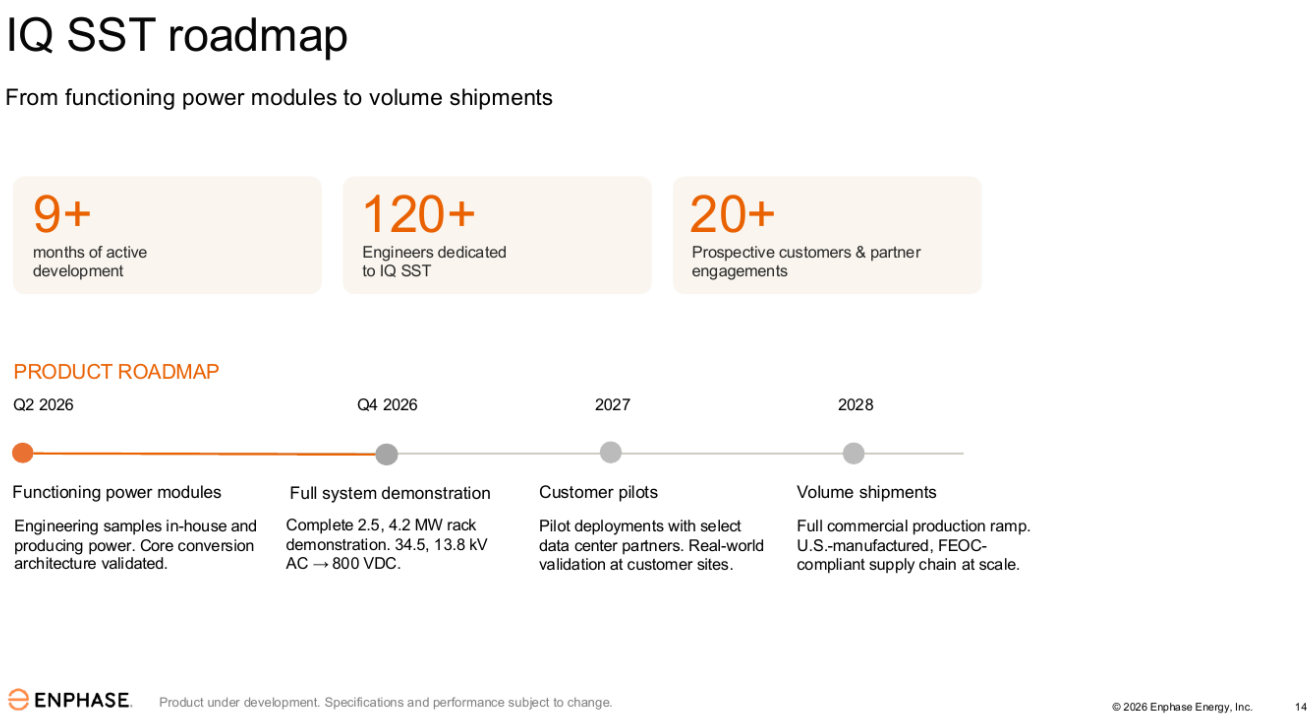
Product demand, costs and conditional SST economics

The reference product model recovers gradually. Microinverter-plus-control unit equivalents rise from 6.8 million in 2027 to 9.8 million in 2030, with blended revenue per unit falling from $145 to $135. Batteries rise from 750 to 1,450 MWh, while revenue per kWh falls from $450 to $395. These are modeling proxies calibrated to reported shipments and consolidated sales; Enphase does not report these product margins or exact blended ASPs. Commercial products and committed PWT deliveries are inside these units.

Underlying microinverter/controls gross margin, before credits, rises from approximately 20.5% in the Q3 guidance calibration to 25% in 2027 and 28% in 2030. Battery gross margin moves from 28% to 32% and eventually 36%. Better product cost and utilization have to earn those gains despite lower prices. R&D, sales/marketing and G&A total $512 million in 2027, $548 million in 2028, $592 million in 2029 and $632 million in 2030. This retains the existing SST team, ordinary amortization and a substantial continuing compensation burden.

For the incremental storage opportunity, a 60% probability applies to 150 MWh of additional 2027 gross demand. Half overlaps the growing reference. The remaining 75 MWh at $450/kWh produces $33.75 million conditional revenue. A 32% GM and 10% incremental marketing/service cost produce $7.425 million conditional OP, or $4.455 million probability weighted. This is a company-specific conversion of customer evidence, not a percentage of a global battery market.

The SST commercialization branch has a 45% probability. Conditional accepted 2028 shipments are bounded at 0-500 MW, with 250 MW selected, followed by 1.1 GW in 2029 and 2 GW in 2030. The chosen $0.15/W equipment ASP lies in a $0.10-0.25/W analytical range and is not a company quote. There are no commercial volume sales in 2027. The issuer roadmap remains the anchor for timing.



Roadmap distinguishes 2027 pilots from 2028 commercial volume.

At the selected 2028 outcome, $37.5 million revenue yields 38% GP before tax credits, less 12% variable support and $12 million incremental fixed deployment cost. That is a **$2.25 million conditional operating loss**, even if the first commercial rollout succeeds. Its probability-weighted impact is $16.875 million revenue and a $1.013 million loss. The branch also pays $8 million of conditional incremental 2027 customer-validation expense; core engineering already sits in reference R&D. Later profits improve only with accepted output and a less burdensome deployment-cost ratio. This avoids assigning mature rack margins to early pilot hardware.

A 30% probability of a 10% shortfall in established-product sales captures weak residential financing and price pressure. It loses the associated product GP and manufacturing credits, while only 12% of the lost sales is saved in variable expense. A separate 15% probability of a 20% loss of future net solar credits from 2028 addresses compliance and monetization risk beyond the statutory phase-down. These risks are not zeroed simply because the present quarter's guidance is positive.

Manufacturing credits are profit, but their category and collection matter

Q2 generated $71.133 million of gross manufacturing credits and recorded $64.922 million net, a 91.27% realization after discounts and fees. The reference model separates this COGS benefit from product economics: 90% U.S.-eligible microinverter equivalents, 400 W AC average eligible rating, the 11¢/W rate, and observed net realization. These are calibrated assumptions; reported DC shipment megawatts must not be substituted for rated AC watts. The 2030 reference applies a 25% statutory reduction in the credit rate. June filing, government-grants note

SST eligibility is a different question. The IRS microinverter definition caps rated capacity at 650 W and requires a qualifying product function. The modeled SST module is 4 kW. Management's positive comments deserve weight, but do not establish the 11¢ rate. Conditional on SST deployment, the model assigns a 20% probability to a qualifying central-inverter-equivalent outcome at 0.25¢/W, bounded by a zero-to-1.5¢ category sensitivity. It also applies U.S. eligibility, transfer discounts and 50% sharing through customer pricing. The joint probability is 9%. This small positive increment acknowledges the breadcrumb while avoiding a much larger unsupported subsidy assumption. It is a modeling hypothesis, not a legal conclusion about the product. IRS final component definitions

The active annual model reaches $1.304 billion revenue and $75.9 million OP in 2027, $1.508 billion/$145.7 million in 2028, $1.766 billion/$227.5 million in 2029, and $2.017 billion/$227.5 million in 2030. The roughly flat 2030 profit despite sales growth illustrates credit erosion, price competition and continuing expense. The forecast does not assume an unlimited SST margin or a permanently subsidized core business.

Shares, debt and the ability to fund the plan

The dated legal common count is 132.144648 million on July 20. The economic denominator includes the entire 3.573 million outstanding RSU pool and 2.182 million expected PSU pool, then adds 0.25 million shares each quarter for net new awards. The filed Q2 diluted EPS denominator is different: its 135.122 million includes 2.018 million if-converted note shares. Starting there and deducting the same debt principal would mix two claims conventions. All future SBC remains expensed despite the separate share allowance.

The remaining notes have $575 million principal, no regular coupon and a March 1, 2028 maturity. The model pays principal once in Q1 2028. Conversion at $284.87 and warrants at $397.91 are above all modeled outcomes, so the forecast assumes cash repayment, no conversion shares, no warrant cash and no independent hedge credit. The $632.5 million 2026 notes were already repaid in March and do not recur. Operating leases and warranty service remain in costs, rather than being deducted a second time as debt claims.

June cash and marketable securities totaled $937.708 million. Funding tests all 24 coherent operating combinations, including SST failure and solar weakness, rather than financing only the weighted average. The most constrained path retains $178.6 million. It includes the full debt repayment, capex, additional SST capital, cash taxes and $10 million quarterly award withholding. No future buyback, financing or investment income is required.

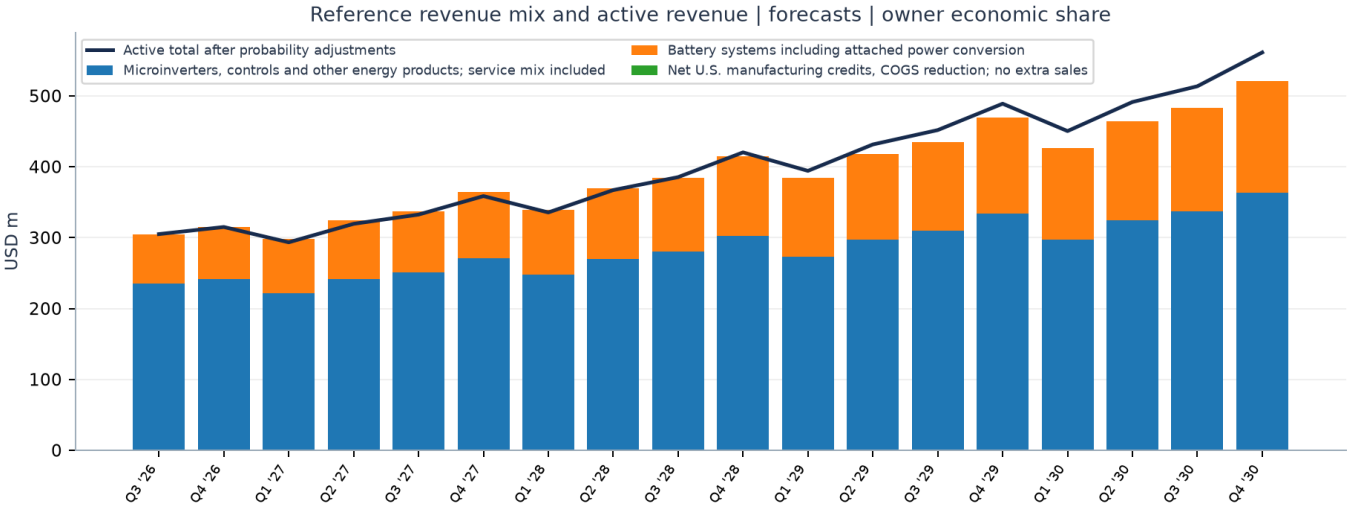
The cash test is deliberately conservative. It assumes no recovery of the old $108.3 million tax-credit receivable and no release of opening noncash operating assets. New credits collect with a two-quarter lag. It reserves the entire remaining $582.505 million deferred-revenue recognition schedule as previously collected cash, without releasing matched contract costs; that is a solvency stress rather than a literal GAAP cash-flow projection. The $41.861 million after-2030 remainder is segregated from available opening cash. This treatment prevents customer deposits or accounting credits from being mistaken for new cash.

The required reassessment points are straightforward: paid pilot economics, medium-voltage qualification, warranty data, new-credit cash conversion, sell-through and the ability to hold product margins after price concessions. Strong evidence on those points can raise either accepted volumes or retained margins before a final reported earnings ramp. A diagram or a broad AI power shortage alone cannot establish those quantities.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	16	Proprietary control silicon and mature distributed-power platform; no proven data-center rack monopoly.
Cycle	5	Residential reset and channel recovery remain fragile.
TAM	8	Storage and SST widen applications, with qualification and wallet limits.
Catalysts	4	G5 launch, full SST demo and paid pilot acceptance.
Macro	7	AI power need and distributed energy support demand; policy/financing are material dependencies.
Alpha	3	Engineering and credit-accounting differentiation are useful but commercialization is unsettled.
Risk classes	-5	Tax-credit reliance, customer concentration, price competition and new-rack qualification.
Total	38/60	Rounded Base 63. Both artifact and new workbook row use this score; this is not a manual override.

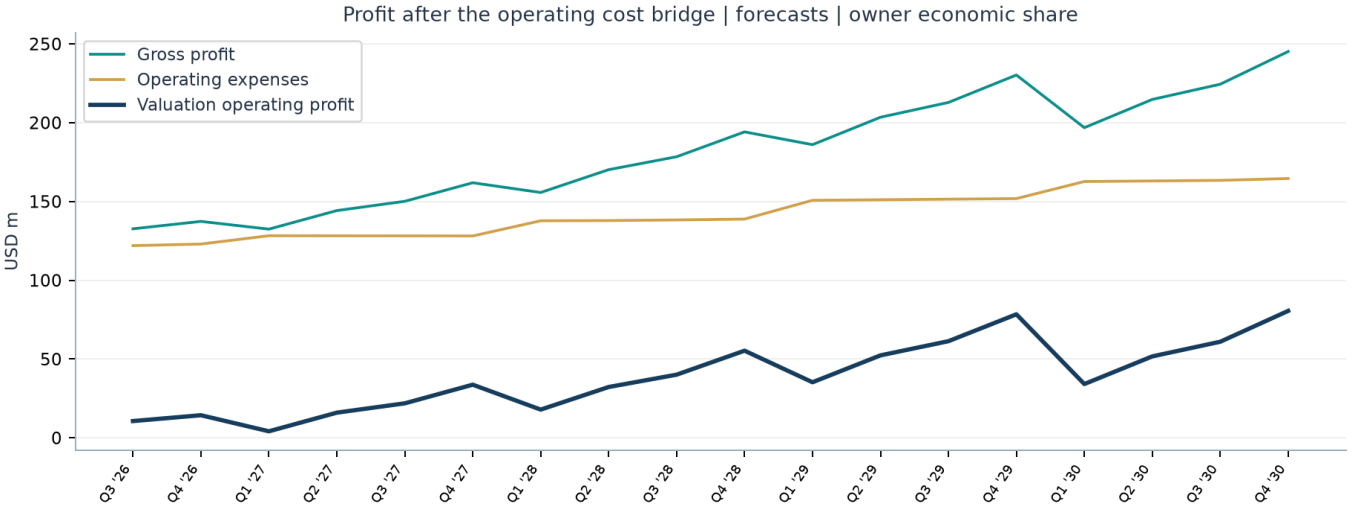
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	305.0	0.0	305.0
Q4 2026	315.0	0.0	315.0
Q1 2027	297.8	-4.4	293.4
Q2 2027	324.3	-4.8	319.5
Q3 2027	337.5	-5.0	332.5
Q4 2027	364.0	-5.4	358.6
Q1 2028	339.4	-3.8	335.6
Q2 2028	369.5	-2.4	367.1
Q3 2028	384.6	0.8	385.4
Q4 2028	414.8	5.5	420.3
Q1 2029	383.9	10.3	394.2
Q2 2029	418.1	13.4	431.4
Q3 2029	435.1	16.6	451.7
Q4 2029	469.3	19.7	489.0
Q1 2030	426.5	23.8	450.3
Q2 2030	464.5	26.9	491.4
Q3 2030	483.4	30.1	513.5
Q4 2030	521.3	40.0	561.3

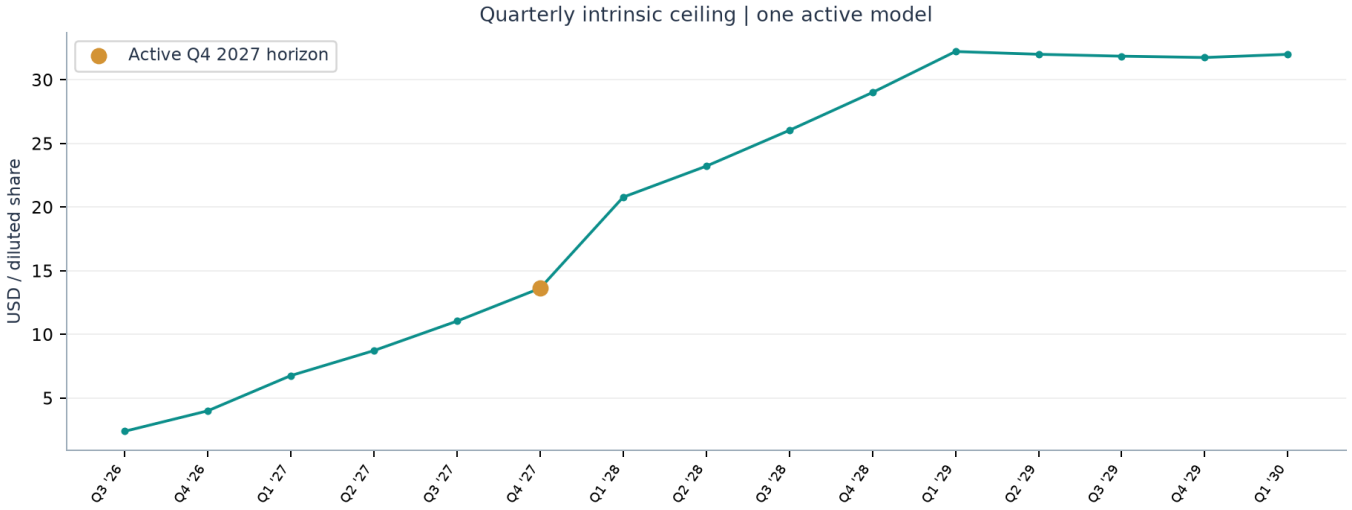
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	305.0	132.7	122.0	10.7	3.5%
Q4 2026	315.0	137.4	123.0	14.4	4.6%
Q1 2027	293.4	132.5	128.3	4.2	1.4%
Q2 2027	319.5	144.2	128.2	16.0	5.0%
Q3 2027	332.5	150.1	128.2	21.9	6.6%
Q4 2027	358.6	161.9	128.1	33.8	9.4%
Q1 2028	335.6	155.7	137.8	17.9	5.3%
Q2 2028	367.1	170.2	137.9	32.3	8.8%
Q3 2028	385.4	178.4	138.3	40.1	10.4%
Q4 2028	420.3	194.2	138.9	55.3	13.2%
Q1 2029	394.2	186.1	150.8	35.3	9.0%
Q2 2029	431.4	203.5	151.1	52.4	12.1%
Q3 2029	451.7	212.9	151.5	61.4	13.6%
Q4 2029	489.0	230.3	151.8	78.4	16.0%
Q1 2030	450.3	196.9	162.7	34.2	7.6%
Q2 2030	491.4	214.8	163.0	51.7	10.5%
Q3 2030	513.5	224.4	163.4	61.0	11.9%
Q4 2030	561.3	245.2	164.6	80.6	14.4%

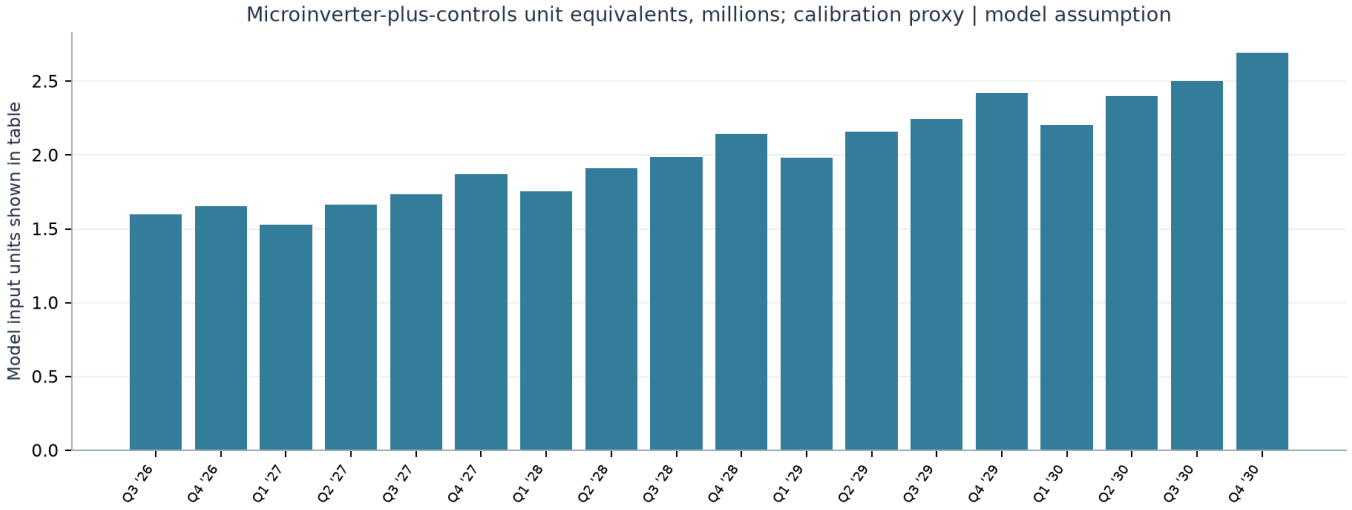
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	45.2	20x	138.650	2.40
Q4 2026	Q4 2026 to Q3 2027	56.5	20x	138.900	4.01
Q1 2027	Q1 2027 to Q4 2027	75.9	20x	139.150	6.77
Q2 2027	Q2 2027 to Q1 2028	89.6	20x	139.400	8.74
Q3 2027	Q3 2027 to Q2 2028	105.9	20x	139.650	11.05
Q4 2027	Q4 2027 to Q3 2028	124.1	20x	139.900	13.63
Q1 2028	Q1 2028 to Q4 2028	145.7	20x	140.150	20.79
Q2 2028	Q2 2028 to Q1 2029	163.0	20x	140.400	23.22
Q3 2028	Q3 2028 to Q2 2029	183.1	20x	140.650	26.04
Q4 2028	Q4 2028 to Q3 2029	204.4	20x	140.900	29.01
Q1 2029	Q1 2029 to Q4 2029	227.5	20x	141.150	32.23
Q2 2029	Q2 2029 to Q1 2030	226.3	20x	141.400	32.01
Q3 2029	Q3 2029 to Q2 2030	225.7	20x	141.650	31.86
Q4 2029	Q4 2029 to Q3 2030	225.3	20x	141.900	31.76
Q1 2030	Q1 2030 to Q4 2030	227.5	20x	142.150	32.01

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	Microinverter-plus-controls unit equivalents, millions; calibration proxy	Status
Q3 2026	1.599	Issuer midpoint calibration
Q4 2026	1.654	Model assumption; not company guidance
Q1 2027	1.530	Model assumption; not company guidance
Q2 2027	1.666	Model assumption; not company guidance
Q3 2027	1.734	Model assumption; not company guidance
Q4 2027	1.870	Model assumption; not company guidance
Q1 2028	1.755	Model assumption; not company guidance
Q2 2028	1.911	Model assumption; not company guidance
Q3 2028	1.989	Model assumption; not company guidance
Q4 2028	2.145	Model assumption; not company guidance
Q1 2029	1.980	Model assumption; not company guidance
Q2 2029	2.156	Model assumption; not company guidance
Q3 2029	2.244	Model assumption; not company guidance
Q4 2029	2.420	Model assumption; not company guidance
Q1 2030	2.205	Model assumption; not company guidance
Q2 2030	2.401	Model assumption; not company guidance
Q3 2030	2.499	Model assumption; not company guidance
Q4 2030	2.695	Model assumption; not company guidance

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Microinverters, controls and other energy products; service mix included	235.0	20.5%	48.1
Q3 2026	Battery systems including attached power conversion	70.0	28.0%	19.6
Q3 2026	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	65.0
Q4 2026	Microinverters, controls and other energy products; service mix included	241.5	23.0%	55.5
Q4 2026	Battery systems including attached power conversion	73.5	30.0%	22.1
Q4 2026	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	59.8
Q1 2027	Microinverters, controls and other energy products; service mix included	221.8	25.0%	55.5
Q1 2027	Battery systems including attached power conversion	75.9	32.0%	24.3
Q1 2027	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	55.3
Q2 2027	Microinverters, controls and other energy products; service mix included	241.6	25.0%	60.4
Q2 2027	Battery systems including attached power conversion	82.7	32.0%	26.5
Q2 2027	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	60.2
Q3 2027	Microinverters, controls and other energy products; service mix included	251.4	25.0%	62.9
Q3 2027	Battery systems including attached power conversion	86.1	32.0%	27.5
Q3 2027	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	62.7
Q4 2027	Microinverters, controls and other energy products; service mix included	271.2	25.0%	67.8
Q4 2027	Battery systems including attached power conversion	92.8	32.0%	29.7
Q4 2027	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	67.6

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2028	Microinverters, controls and other energy products; service mix included	247.5	26.5%	65.6
Q1 2028	Battery systems including attached power conversion	91.9	34.0%	31.3
Q1 2028	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	63.4
Q2 2028	Microinverters, controls and other energy products; service mix included	269.5	26.5%	71.4
Q2 2028	Battery systems including attached power conversion	100.1	34.0%	34.0
Q2 2028	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	69.1
Q3 2028	Microinverters, controls and other energy products; service mix included	280.4	26.5%	74.3
Q3 2028	Battery systems including attached power conversion	104.2	34.0%	35.4
Q3 2028	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	71.9
Q4 2028	Microinverters, controls and other energy products; service mix included	302.4	26.5%	80.1
Q4 2028	Battery systems including attached power conversion	112.3	34.0%	38.2
Q4 2028	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	77.5
Q1 2029	Microinverters, controls and other energy products; service mix included	273.2	27.5%	75.1
Q1 2029	Battery systems including attached power conversion	110.7	35.0%	38.7
Q1 2029	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	71.6
Q2 2029	Microinverters, controls and other energy products; service mix included	297.5	27.5%	81.8
Q2 2029	Battery systems including attached power conversion	120.5	35.0%	42.2
Q2 2029	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	77.9
Q3 2029	Microinverters, controls and other energy products; service mix included	309.7	27.5%	85.2
Q3 2029	Battery systems including attached power conversion	125.5	35.0%	43.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2029	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	81.1
Q4 2029	Microinverters, controls and other energy products; service mix included	334.0	27.5%	91.8
Q4 2029	Battery systems including attached power conversion	135.3	35.0%	47.4
Q4 2029	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	87.5
Q1 2030	Microinverters, controls and other energy products; service mix included	297.7	28.0%	83.3
Q1 2030	Battery systems including attached power conversion	128.9	36.0%	46.4
Q1 2030	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	59.8
Q2 2030	Microinverters, controls and other energy products; service mix included	324.1	28.0%	90.8
Q2 2030	Battery systems including attached power conversion	140.3	36.0%	50.5
Q2 2030	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	65.1
Q3 2030	Microinverters, controls and other energy products; service mix included	337.4	28.0%	94.5
Q3 2030	Battery systems including attached power conversion	146.1	36.0%	52.6
Q3 2030	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	67.7
Q4 2030	Microinverters, controls and other energy products; service mix included	363.8	28.0%	101.9
Q4 2030	Battery systems including attached power conversion	157.5	36.0%	56.7
Q4 2030	Net U.S. manufacturing credits, COGS reduction; no extra sales	0.0	0.0%	73.1

Operating-cost derivation

Quarter	R&D including SST development, SBC and depreciation	Sales and marketing including SBC	General and administrative including SBC and amortization
Q3 2026	46.0	45.0	31.0
Q4 2026	47.0	45.0	31.0
Q1 2027	50.0	46.2	31.8
Q2 2027	50.0	46.2	31.8
Q3 2027	50.0	46.2	31.8
Q4 2027	50.0	46.2	31.8
Q1 2028	53.8	49.5	33.8
Q2 2028	53.8	49.5	33.8
Q3 2028	53.8	49.5	33.8
Q4 2028	53.8	49.5	33.8
Q1 2029	58.8	53.8	35.5
Q2 2029	58.8	53.8	35.5
Q3 2029	58.8	53.8	35.5
Q4 2029	58.8	53.8	35.5
Q1 2030	63.8	56.8	37.5
Q2 2030	63.8	56.8	37.5
Q3 2030	63.8	56.8	37.5
Q4 2030	63.8	56.8	37.5

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	ENPH-BATTERY	60%	0.0	0.0	0.0	0.0
Q3 2026	ENPH-SST	45%	0.0	0.0	0.0	0.0
Q3 2026	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q3 2026	ENPH-SOLAR-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q3 2026	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q4 2026	ENPH-BATTERY	60%	0.0	0.0	0.0	0.0
Q4 2026	ENPH-SST	45%	0.0	0.0	0.0	0.0
Q4 2026	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q4 2026	ENPH-SOLAR-DOWNSIDE	30%	0.0	0.0	0.0	0.0
Q4 2026	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q1 2027	ENPH-BATTERY	60%	7.6	2.4	0.8	1.0
Q1 2027	ENPH-SST	45%	0.0	0.0	2.0	-0.9
Q1 2027	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q1 2027	ENPH-SOLAR-DOWNSIDE	30%	-29.8	-13.5	-3.6	-3.0
Q1 2027	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q2 2027	ENPH-BATTERY	60%	8.3	2.6	0.8	1.1
Q2 2027	ENPH-SST	45%	0.0	0.0	2.0	-0.9
Q2 2027	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q2 2027	ENPH-SOLAR-DOWNSIDE	30%	-32.4	-14.7	-3.9	-3.2
Q2 2027	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q3 2027	ENPH-BATTERY	60%	8.6	2.8	0.9	1.1
Q3 2027	ENPH-SST	45%	0.0	0.0	2.0	-0.9
Q3 2027	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q3 2027	ENPH-SOLAR-DOWNSIDE	30%	-33.7	-15.3	-4.0	-3.4
Q3 2027	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q4 2027	ENPH-BATTERY	60%	9.3	3.0	0.9	1.2

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q4 2027	ENPH-SST	45%	0.0	0.0	2.0	-0.9
Q4 2027	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q4 2027	ENPH-SOLAR-DOWNSIDE	30%	-36.4	-16.5	-4.4	-3.6
Q4 2027	ENPH-CREDIT-RISK	15%	0.0	0.0	0.0	0.0
Q1 2028	ENPH-BATTERY	60%	10.6	3.6	1.1	1.5
Q1 2028	ENPH-SST	45%	0.0	0.0	3.0	-1.4
Q1 2028	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q1 2028	ENPH-SOLAR-DOWNSIDE	30%	-33.9	-16.0	-4.1	-3.6
Q1 2028	ENPH-CREDIT-RISK	15%	0.0	-12.7	0.0	-1.9
Q2 2028	ENPH-BATTERY	60%	11.6	3.9	1.2	1.7
Q2 2028	ENPH-SST	45%	3.8	1.4	3.5	-0.9
Q2 2028	ENPH-SST-CREDIT	9%	0.0	0.0	0.0	0.0
Q2 2028	ENPH-SOLAR-DOWNSIDE	30%	-37.0	-17.5	-4.4	-3.9
Q2 2028	ENPH-CREDIT-RISK	15%	0.0	-13.8	0.0	-2.1
Q3 2028	ENPH-BATTERY	60%	12.1	4.1	1.2	1.7
Q3 2028	ENPH-SST	45%	11.2	4.3	4.3	-0.0
Q3 2028	ENPH-SST-CREDIT	9%	0.0	0.1	0.0	0.0
Q3 2028	ENPH-SOLAR-DOWNSIDE	30%	-38.5	-18.2	-4.6	-4.1
Q3 2028	ENPH-CREDIT-RISK	15%	0.0	-14.4	0.0	-2.2
Q4 2028	ENPH-BATTERY	60%	13.0	4.4	1.3	1.9
Q4 2028	ENPH-SST	45%	22.5	8.6	5.7	1.3
Q4 2028	ENPH-SST-CREDIT	9%	0.0	0.2	0.0	0.0
Q4 2028	ENPH-SOLAR-DOWNSIDE	30%	-41.5	-19.6	-5.0	-4.4
Q4 2028	ENPH-CREDIT-RISK	15%	0.0	-15.5	0.0	-2.3
Q1 2029	ENPH-BATTERY	60%	13.8	4.8	1.4	2.1
Q1 2029	ENPH-SST	45%	30.0	12.0	7.3	2.1
Q1 2029	ENPH-SST-CREDIT	9%	0.0	0.2	0.0	0.0
Q1 2029	ENPH-SOLAR-DOWNSIDE	30%	-38.4	-18.5	-4.6	-4.2

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q1 2029	ENPH-CREDIT-RISK	15%	0.0	-14.3	0.0	-2.1
Q2 2029	ENPH-BATTERY	60%	15.1	5.3	1.5	2.3
Q2 2029	ENPH-SST	45%	37.5	15.0	8.2	3.0
Q2 2029	ENPH-SST-CREDIT	9%	0.0	0.3	0.0	0.0
Q2 2029	ENPH-SOLAR-DOWNSIDE	30%	-41.8	-20.2	-5.0	-4.6
Q2 2029	ENPH-CREDIT-RISK	15%	0.0	-15.6	0.0	-2.3
Q3 2029	ENPH-BATTERY	60%	15.7	5.5	1.6	2.4
Q3 2029	ENPH-SST	45%	45.0	18.0	9.1	4.0
Q3 2029	ENPH-SST-CREDIT	9%	0.0	0.3	0.0	0.0
Q3 2029	ENPH-SOLAR-DOWNSIDE	30%	-43.5	-21.0	-5.2	-4.7
Q3 2029	ENPH-CREDIT-RISK	15%	0.0	-16.2	0.0	-2.4
Q4 2029	ENPH-BATTERY	60%	16.9	5.9	1.7	2.5
Q4 2029	ENPH-SST	45%	52.5	21.0	10.1	4.9
Q4 2029	ENPH-SST-CREDIT	9%	0.0	0.4	0.0	0.0
Q4 2029	ENPH-SOLAR-DOWNSIDE	30%	-46.9	-22.7	-5.6	-5.1
Q4 2029	ENPH-CREDIT-RISK	15%	0.0	-17.5	0.0	-2.6
Q1 2030	ENPH-BATTERY	60%	16.0	5.8	1.6	2.5
Q1 2030	ENPH-SST	45%	60.0	25.2	11.7	6.1
Q1 2030	ENPH-SST-CREDIT	9%	0.0	0.3	0.0	0.0
Q1 2030	ENPH-SOLAR-DOWNSIDE	30%	-42.7	-19.0	-5.1	-4.1
Q1 2030	ENPH-CREDIT-RISK	15%	0.0	-12.0	0.0	-1.8
Q2 2030	ENPH-BATTERY	60%	17.4	6.3	1.7	2.7
Q2 2030	ENPH-SST	45%	67.5	28.3	12.6	7.1
Q2 2030	ENPH-SST-CREDIT	9%	0.0	0.3	0.0	0.0
Q2 2030	ENPH-SOLAR-DOWNSIDE	30%	-46.4	-20.6	-5.6	-4.5
Q2 2030	ENPH-CREDIT-RISK	15%	0.0	-13.0	0.0	-2.0
Q3 2030	ENPH-BATTERY	60%	18.1	6.5	1.8	2.8
Q3 2030	ENPH-SST	45%	75.0	31.5	13.5	8.1

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2030	ENPH-SST-CREDIT	9%	0.0	0.4	0.0	0.0
Q3 2030	ENPH-SOLAR-DOWNSIDE	30%	-48.3	-21.5	-5.8	-4.7
Q3 2030	ENPH-CREDIT-RISK	15%	0.0	-13.5	0.0	-2.0
Q4 2030	ENPH-BATTERY	60%	19.6	7.0	2.0	3.1
Q4 2030	ENPH-SST	45%	97.5	40.9	16.2	11.1
Q4 2030	ENPH-SST-CREDIT	9%	0.0	0.5	0.0	0.0
Q4 2030	ENPH-SOLAR-DOWNSIDE	30%	-52.1	-23.2	-6.3	-5.1
Q4 2030	ENPH-CREDIT-RISK	15%	0.0	-14.6	0.0	-2.2

Enterprise-to-equity reconciliation

20x forward-four-quarter normalized GAAP operating profit including SBC and D&A, less outstanding note principal at labeled quarter end, divided by end-window economic shares. No cash addition. Quarter includes itself plus next three. Apply the common-equity floor inside each of 24 operating outcomes before weighting; all outcomes are cash-funded.

Quarter	Reference claims	Weighted change	Total claims	Ceiling / share
Q3 2026	575.0	0.0	575.0	2.40
Q4 2026	575.0	0.0	575.0	4.01
Q1 2027	575.0	0.0	575.0	6.77
Q2 2027	575.0	0.0	575.0	8.74
Q3 2027	575.0	0.0	575.0	11.05
Q4 2027	575.0	0.0	575.0	13.63
Q1 2028	0.0	0.0	0.0	20.79
Q2 2028	0.0	0.0	0.0	23.22
Q3 2028	0.0	0.0	0.0	26.04
Q4 2028	0.0	0.0	0.0	29.01
Q1 2029	0.0	0.0	0.0	32.23
Q2 2029	0.0	0.0	0.0	32.01
Q3 2029	0.0	0.0	0.0	31.86
Q4 2029	0.0	0.0	0.0	31.76
Q1 2030	0.0	0.0	0.0	32.01
Q2 2030	0.0	0.0	0.0	-
Q3 2030	0.0	0.0	0.0	-
Q4 2030	0.0	0.0	0.0	-

Coherent Enphase outcomes and funding

Battery, SST commercialization, conditional SST credit, solar weakness and credit risk produce 24 coherent outcomes. SST credits cannot occur without commercialization. Each outcome is floored at common equity before weighting. Cash is reserved for funding; it is not a second valuation asset.

Q4 2027 state bits	Probability	F4Q OP	End-window shares	Ceiling	Minimum cash
00000	13.09%	142.5	139.900	16.25	246.8
01000	8.57%	135.4	139.900	15.24	216.8
01100	2.14%	135.5	139.900	15.25	216.8
00001	2.31%	101.6	139.900	10.41	246.9
01001	1.51%	94.5	139.900	9.40	216.8
01101	0.38%	94.6	139.900	9.41	216.8
00010	5.61%	91.8	139.900	9.01	230.9
01010	3.67%	84.7	139.900	8.00	195.2
01110	0.92%	84.8	139.900	8.01	195.2
00011	0.99%	50.9	139.900	3.17	230.9
01011	0.65%	43.8	139.900	2.15	178.6
01111	0.16%	43.9	139.900	2.17	178.6
10000	19.63%	152.7	139.900	17.72	247.9
11000	12.85%	145.6	139.900	16.71	218.5
11100	3.21%	145.7	139.900	16.72	218.5
10001	3.46%	111.8	139.900	11.88	248.6
11001	2.27%	104.7	139.900	10.86	218.5
11101	0.57%	104.9	139.900	10.88	218.5
10010	8.42%	102.1	139.900	10.48	233.0
11010	5.51%	95.0	139.900	9.47	198.8
11110	1.38%	95.1	139.900	9.48	198.8
10011	1.49%	61.2	139.900	4.64	233.0
11011	0.97%	54.1	139.900	3.62	187.2
11111	0.24%	54.2	139.900	3.64	187.2

Bit order: battery, SST, SST credit, solar weakness, credit risk. A 1 in the last two positions is an adverse outcome. Net manufacturing credits appear as zero-revenue gross-profit benefits in the product schedule, not as sales or a meaningful gross-margin percentage.

Claim decisions, overlap and triggers

ENPH-BATTERY

probability: 0.6

source: Q1/Q2 management calls: G5 cost reduction, meter-collar reuse, European retrofit leads and commercial IQ Vault launch.

Conditional bound: Above the reference 750/950/1,200/1,450 MWh battery path, gross additional volume is 100-200 MWh in 2027, selected 150; 220/300/360 MWh in later years. Remove 50% for reference overlap. Selected battery ASPs $450/$430/$410/$395 per kWh are model proxies, not disclosed prices.

bridge: Net 2027 conditional revenue $33.75m at 32% GAAP gross margin less 10% marketing/service cost: $7.425m OP. Probability-weighted $20.25m revenue and $4.455m OP. Battery price cuts and cell cost reductions are inside these margins.

overlap: Reference already includes G5 and IQ Vault growth. No second addition for safe-harbor attachments, Propel, tariff refunds or battery capacity.

triggers: Accepted volume, installed-base conversion and retained unit margins; reduce on channel restocking without sell-through.

ENPH-SST

probability: 0.45

source: July deck and Q2 call; September 8 issuer: full-rack validation, November demo, 2027 pilots and 2028 commercial timing.

Conditional bound: 2028 accepted capacity 0-500 MW, selected 250 MW; quarterly 0/25/75/150. Selected 1.1 GW in 2029, 2 GW in 2030; no volume revenue in 2027. ASP $0.10-0.25/W, selected $0.15/W, is an analytical hardware-wallet proxy, not a quote/order.

bridge: Conditional 2028 revenue $37.5m, GM 38%, 12% variable support cost and $12m fixed incremental deployment cost gives -$2.25m OP. Weighted $16.875m revenue and -$1.0125m OP. 2027 adds $8m conditional validation/customer cost (-$3.6m weighted OP); reference R&D already pays the core engineering team. GM reaches 40%/42% as accepted output scales; extra capex is explicit.

overlap: One rack wallet only. No double count for a possible BESS second SST, generic DC power TAM, utility applications, pre-existing 12m microinverter capacity, or every module inside a rack.

triggers: Full medium-voltage rack reliability, partial discharge, thermal qualification, accepted customer pilots, paid orders, warranty economics and manufacturing yield. Delay revenue on missed customer qualification.

ENPH-SST-CREDIT

probability: 0.09

Conditional probability given SST: 0.2

source: Q2 management says tax-credit details are being finalized. IRS final regulations define inverter suitability and separate credit categories.

Conditional bound: Conditional on commercial SST deployment, assign 20% probability of a lawful eligible central-inverter-equivalent outcome, selected $0.0025/W, within $0-0.015/W category sensitivity. Multiply accepted MW by 90% U.S. eligibility, current 91.268% monetization and 50% retention after customer price sharing. Joint probability .45*.20=.09; 2030 statutory phase-down is 75%.

bridge: Tax benefit reduces COGS; no revenue. Selected 2028 conditional GP is about $0.257m before joint probability. It is small because a 4 kW module does not satisfy the <=650 W microinverter capacity condition; $0.11/W is not imported.

overlap: Nested under the same accepted SST units, not an independent volume event. The margin assumption excludes credits before this separate branch.

triggers: A documented qualifying product classification, as-sold solar/DC-to-AC suitability, rated AC capacity and transfer economics. This is a bounded modeling hypothesis, not a tax conclusion.

ENPH-SOLAR-DOWNSIDE

probability: 0.3

source: U.S. Q2 sell-through down 34% year over year, price reductions, customer concentration and uncertain third-party-owner funding.

Conditional bound: 5-15% downside to reference energy-product sales from 2027; selected 10%, applied once to both core vectors.

bridge: Lose their corresponding underlying GP and the associated 10% solar credits; save only 12% of lost revenue in variable operating expenses. Fixed engineering/support and SBC remain. No extra tariff charge layered onto guided margins.

overlap: One combined demand/price-financing branch; not a separate charge for each weak distributor, policy headline and interest-rate signal.

triggers: End-customer installations, financing access, channel inventory and price/cost retention.

ENPH-CREDIT-RISK

probability: 0.15

source: 45X monetization discounts, compliance and tax-credit receivable uncertainty in the 10-Q.

Conditional bound: An additional 10-30% reduction in future net solar credits from 2028, selected 20%, beyond the explicitly modeled 2030 statutory 25% phase-down.

bridge: Dollar-for-dollar lower GP/OP; cash flow removes the same credit, with each disjoint modeled loss expressed against reference credit. No automatic replacement equity is required under tested cash paths.

overlap: Does not charge the old 2025 monetization loss again; collection timing reserve is separated from permanent value impairment.

triggers: Qualified U.S. output, FEOC compliance, transfer discounts and tax processing evidence.

Remaining bounded uncertainties

- Product-vector gross margins, blended microinverter/controls ASPs and future battery ASPs are estimates calibrated to consolidated guidance, not reported segment data. The 45X credit is a separate COGS reduction and is not sales.

- Q2 reported OP $51.519m includes the full $45.4m tariff refund. Recurring starting OP is $6.119m; the non-GAAP reconciliation removes $43.4m prior-period refund and is not the same basis. All SBC, D&A and ordinary warranty costs remain in the model.

- 2027-2030 revenue, gross margins and expenses are analytical forecasts, not management guidance. Safe-harbor/PWT commitments are nested within unit sales, never stacked on top.

- SST prices, accepted customer capacity, qualified output, production yield, warranties and customer identities are not disclosed. Nominal microinverter capacity is not proven SST capacity.

- The July deck 2.5 MW/672-module revision differs from May 1.25 MW/342 modules. 672 times 4 kW is 2.688 MW; 90% is only 2.4192 MW, so do not simultaneously promise the rated 2.5 MW at an assumed 10% module loss without the final rating convention.

- The whitepaper compares sub-millisecond response with tens of milliseconds but also describes roughly 1,000 times; the arithmetic is not established from those bounds. Use the absolute response target, not that ratio. Peak efficiency, full-rack efficiency, uptime, insulation qualification and lifetime failure rates are distinct.

- SST 45X classification is unresolved. A small positive central-inverter-equivalent branch expresses management evidence; the microinverter 11-cent rate is excluded by the capacity/function mismatch, not simply because confirmation is missing.

- Funding is a conservative solvency stress, not a clean GAAP cash-flow forecast: existing noncash assets have no release, the old $108.3m credit receives no cash recovery, and the full existing $582.505m deferred-revenue recognition schedule is reserved as previously collected cash without releasing matched contract costs. The $41.861m after-2030 amount is held outside available opening cash. New ordinary trade working capital is 70% of incremental quarterly sales; new credits collect with two-quarter lag.

- Future taxes are a $5m quarterly floor or 22% positive OP, plus $30m near-term catch-up. Capex, $10m quarterly award withholding and incremental SST deployment capital are deducted. No new financing, buybacks or interest-income funding is assumed.

- June 2028-note principal $575m is repaid once in Q1 2028. Valuation deducts debt outstanding at the labeled quarter end, not F4Q end; future cash repayment removes the claim thereafter. Cash is not separately added. Convertible/warrant strikes $284.87/$397.91 are above all model values, so no conversion, exercise cash or hedge credit is assumed.

- Shares start from July20 legal 132.144648m plus all 3.573m RSUs and 2.182m expected PSUs, then rise 0.25m per quarter for net new awards. This is a conservative economic denominator, not the filed EPS denominator, which includes 2.018m if-converted note shares.

- The full Q1/Q2 call text is a third-party reproduction of management discussion. Original issuer financial filings and technical PDFs are archived. September8 release was read through the primary web page; direct local retrieval timed out.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

[1] Enphase 2025 Form 10-K | 2026-02-17 | https://www.sec.gov/Archives/edgar/data/1463101/000146310126000013/enph-20251231.htm | Archived SHA-256: adf01a38efeaf43073f9124144899a2f093f703fde8dff6292fbb0e668a2a67b

[2] Enphase 2025 annual report PDF | 2026-03-26 | https://www.sec.gov/Archives/edgar/data/1463101/000146310126000030/annualreport2025.pdf | Archived SHA-256: a2d95a0e48be5621e2f3e37e78c7812982f9fb4cb745aee0b3d336f610bf7fc3

[3] Enphase June 2026 Form 10-Q | 2026-07-28 | https://www.sec.gov/Archives/edgar/data/1463101/000146310126000081/enph-20260630.htm | Archived SHA-256: 0d75b2ef93f0aa018a8ba165f2a0cd375663fbdf3018cc19ea20fc59b55f0981

[4] Enphase Q2 2026 earnings release | 2026-07-28 | https://www.sec.gov/Archives/edgar/data/1463101/000146310126000080/a2026q2exx991pressrelease.htm | Archived SHA-256: 8eb18ab7e1829a9b36e2c19b9c37d40dc199f75f07fd4751162d3d16dac00890

[5] Enphase Q1 2026 earnings release | 2026-04-28 | https://www.sec.gov/Archives/edgar/data/1463101/000146310126000046/a2026q1exx991pressrelease.htm | Archived SHA-256: 578d61b3b5db22d6924e74c6e28576e4abc918e0e15de5f719bd7a71c0ccf22e

[6] Q2 2026 management call, Quartr transcription | 2026-07-28 | https://stockanalysis.com/stocks/enph/transcripts/656436-q2-2026/ | Archived SHA-256: 23b5a939c60edbb7bef91300ac82498b73f89d907d48a5dbadf7c484e1b1ceaa

[7] Q1 2026 management call, Quartr transcription | 2026-04-28 | https://stockanalysis.com/stocks/enph/transcripts/551243-q1-2026/ | Archived SHA-256: c538b71b1fe2f6347965dca91efaa63584ae6e4a0b92219672856210a6f3ed8d

[8] Enphase SST investor deck, July 30 file revision | 2026-07 | https://enphase.com/download/enphase-sst-investor-deck | Archived SHA-256: 89af34b78f32c467682fb6f4423b1a299d04e3e8a0d011ca24e1b168df4ec2a9

[9] Kestrel ASIC whitepaper, June 2026 / published July 27 | 2026-07-27 | https://enphase.com/download/kestrel-asic-technology-white-paper | Archived SHA-256: 2fff21b88d456a7a0c233f5b9a9fbdc1d9b5785fee2bc0adc2de67edcf939637

[10] GaN bidirectional-switch whitepaper, May 2026 | 2026-05-21 | https://enphase.com/download/enphase-energy-adoption-gan-bds | Archived SHA-256: 7a81a626909829d4d6621a547684357c4b32c5e6eb85aaff7ca4e05caeb2b6a1

[11] IRS final 45X component regulations, December 2024 | 2024-12-16 | https://www.irs.gov/irb/2024-51_IRB | Archived SHA-256: ab658b577cad32ece06f3cc5a92e1593e721cc5edd53cb15a8e224dc91ef549d

[12] Texas power-module production for full-rack validation | 2026-09-08 | https://newsroom.enphase.com/news-releases/news-release-details/enphase-advances-iq-solid-state-transformer-development-ai-data